Project Proposal

EE322: Analog and Mixed Signal Circuits

Team Name: Team Brownian

Team Members:

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Project Proposal

Introduction

This project focuses on creating a simple all-analog battery charger that takes full control of the charging process without needing any complex digital devices or microcontroller. By using only basic analog components like transistors, op amps, and MOSFETs, the charger automatically moves through the key charging stages-constant current, constant voltage, and trickle charge. This ensures the battery gets charged quickly, safely, and without overcharging or damage. The compact and straightforward design promotes a deep understanding of analog control principles and makes the circuit reliable, affordable, and easy to build.

Project Statement

This project focuses on developing a straightforward analog charger for 12 V lead-acid batteries that automatically manages charging through three essential phases: constant current, constant voltage, and trickle charge. To help users monitor the charging process, visual LED indicators are integrated. The charger features a low-dropout design, which maintains proper voltage regulation even when the input voltage only slightly exceeds the battery voltage, enhancing overall efficiency.

Proposed Approach

Overview

The proposed approach for this problem statement is to build a low-dropout CC-CV (Constant Current - Constant Voltage) charger with trickle charge mode, all controlled by analog feedback system. A P-channel MOSFET acts as the main power-pass element. It will achieve automatic switching between Constant Current and Constant Voltage, and trickle when the battery is charged fully, to prevent any damages. Constant current charging charges the initially discharged battery in bulk. When the voltage reaches its threshold voltage, it will automatically switch to Constant Voltage mode. Finally, once the battery is charged, it will switch to the trickle mode to prevent overcharging and discharging.

Components and Blocks

Our design circuit is broken down into 7 blocks primarily.

Full System Diagram (Block Diagram)

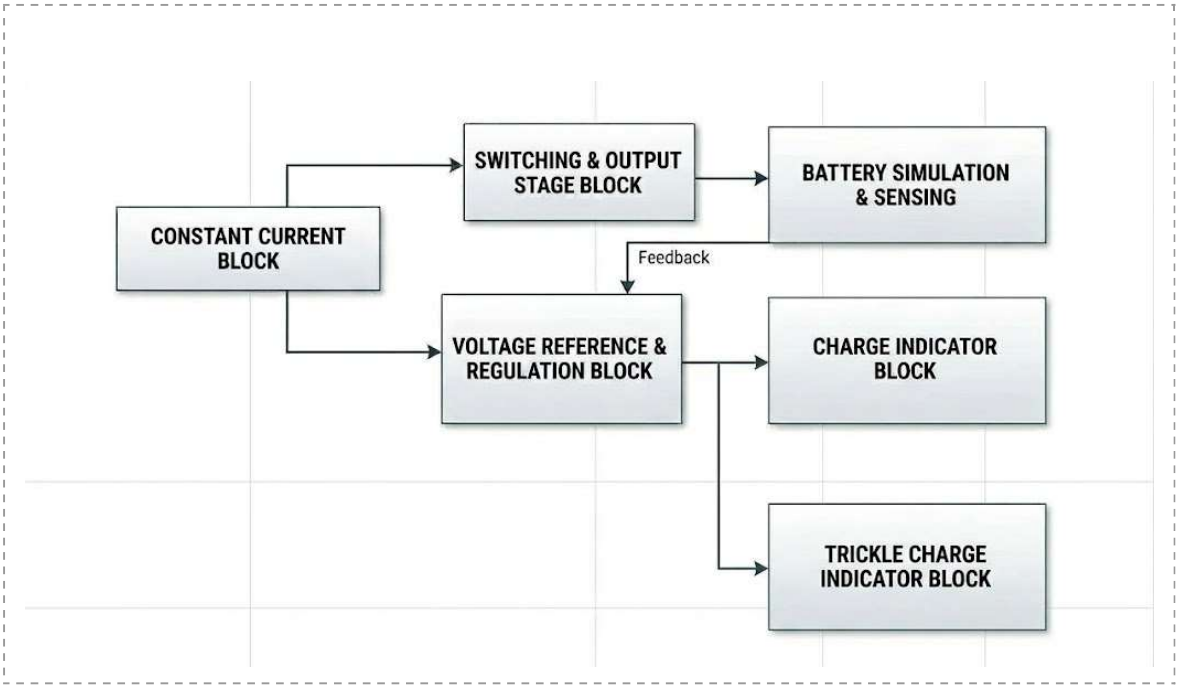


Figure 1: Full System Block Diagram

Group PCB Planning

Team Member's Name	Block No.	Block Name
Amrit	2	Voltage Reference and Regulation Block
Amrit	3	Constant Current Block
Akhil	4	Switching and Output Stage Block
Akhil	5	Battery Simulation and Sensing
Rahul	6	Charge Indicator Block
Rahul	7	Trickle Charge Indicator Block

Block Descriptions by Team Member

Amrit's Blocks (Control & Feedback Logic)

Block 2: Voltage Reference and Regulation Block

- **Function:** Monitors battery voltage to prevent overcharging.
- **Primary Components:** TL431A Programmable Voltage Reference, 2N3906 transistor.
- **Interaction:** Senses output voltage; sends control signals to throttle power.
- **Expected Outcome:** Accurate voltage setpoints maintained and controlled output voltage.

Block 3: Constant Current Block

- **Function:** Controls bulk charging current to safe preset limits.
- **Primary Components:** TL431A and 2N3904 transistors.
- **Interaction:** Receives current sense feedback; sends control signals to limit power.
- **Expected Outcome:** Charging current limited to desired level during bulk charge phase.

Akhil's Blocks (Power Switching & Output Stage)

Block 4: Switching and Output Stage Block

- **Function:** Main MOSFET acting as an electronic valve regulating current flow.
- **Primary Components:** AO6407 P-channel MOSFET.
- **Interaction:** Receives control signals to regulate current to battery.
- **Expected Outcome:** Efficient current regulation; safe operation with proper heat dissipation.

Block 5: Battery Simulation and Sensing

- **Function:** Low-ohm shunt resistor for current sensing; connection point for battery.
- **Primary Components:** 1Ω high-power sense resistor (Rlim).
- **Interaction:** Senses charging current; feedback to Constant Current block.

- **Expected Outcome:** Secure battery connection and accurate sensing voltage.

Rahul's Block (Charge and Trickle Charge Indicator)

Blocks 6 & 7: Charge & Trickle Charge Indicator Blocks

- **Function:** Provide visual feedback via LEDs on charger status.
- **Primary Components:** Indicator LEDs, 2N7000 N-channel MOSFETs.
- **Interaction:** Receive control signals from logic blocks; act as output indicators.
- **Expected Outcome:**
  - 'Charge Indicator' LED on during Constant Current and high-current Constant Voltage phases.
  - 'Trickle Charge Indicator' LED on during final maintenance phase.

Plan for Full Integration

Each step should be completed correctly before moving to the next:

1. Test the Voltage Reference and Regulation Block together with the Constant Current Block on a breadboard using a lab power supply to get their setpoints.
2. Perform a basic function test on the Switching and Output Stage Block to ensure the MOSFET is working.
3. Connect the control logic blocks to the Switching and Output Stage Block and conduct initial testing using a high-power dummy load. It should demonstrate stable Constant Current (CC) and Constant Voltage (CV) regulation.
4. Connect the Charge Indicator Block and the Trickle Charge Indicator Block to the control logic and test by simulating different charging phases to ensure the correct LEDs light up for each mode.
5. Connect the fully integrated and tested charger to a partially discharged 12 V lead-acid battery. The charger must automatically transition through all three stages (CC, CV, and Trickle Charge) and bring the battery to full charge.

Advantages of the Proposed Approach

The proposed all-analog solution offers several significant benefits compared to simple trickle chargers or microcontroller-based systems:

- Highly reliable and free from code bugs.
- Extends battery life by preventing overcharging, gassing, and overheating.
- Matched transistor pairs ensure precision and stability across temperature variations.
- Significantly faster charging than simple trickle chargers.
- Less complex than microcontroller solutions, enhancing accuracy.
- Easy to modify and adapt to specific needs.

Working of the Circuit

Schematic

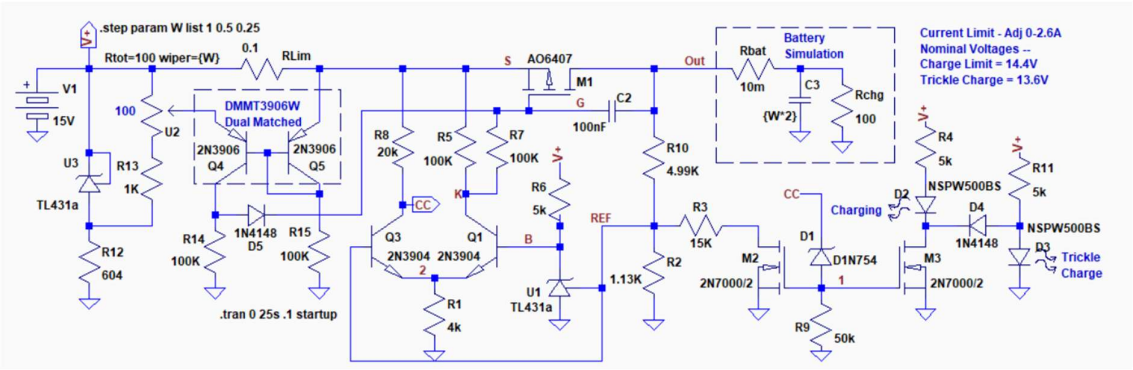


Figure 3: Schematic

Step-by-Step Approach

References

- **A Low Dropout CC-CV 12V Lead Acid Battery Charger with Auto Trickle Charge, EE World Online,** Available: https://www.eeworldonline.com/a-low-dropout-cc-cv-12v-lead-acid-battery-charger-with-auto-trickle-charge/